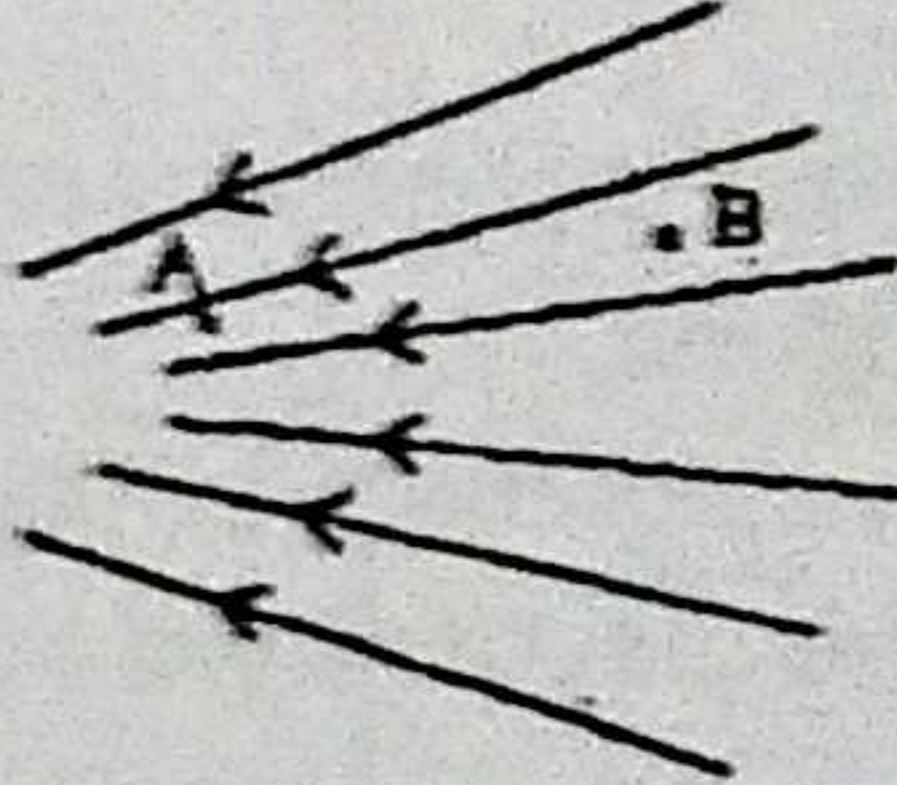


Electrostatics CPP - II

1. Charge Q is distributed non-uniformly over a ring of radius R , P is a point on the axis of ring at a distance $\sqrt{3}R$ from its centre. Which of the following is a wrong statement?
 - (a) Potential at P is $\frac{KQ}{2R}$
 - (b) Magnitude of electric field at P may be greater than $\frac{\sqrt{3}KQ}{8R^2}$
 - (c) Magnitude of electric field at P must be equal to $\frac{\sqrt{3}KQ}{8R^2}$
 - (d) Magnitude of electric field at P cannot be less than $\frac{\sqrt{3}KQ}{8R^2}$
2. An electric dipole moment $\vec{p} = (2.0\hat{i} + 3.0\hat{j}) \mu\text{C} \cdot \text{m}$ is placed in a uniform electric field $\vec{E} = (3.0\hat{i} + 2.0\hat{k}) \times 10^5 \text{ NC}^{-1}$.
 - (a) The torque that $\vec{E}$ exerts on $\vec{p}$ $(0.6\hat{i} - 0.4\hat{j} - 0.9\hat{k}) \text{ Nm}$.
 - (b) The potential energy of the dipole is -0.6 J
 - (c) The potential energy of the dipole is 0.6 J
 - (d) If the dipole is rotated in the electric field, the maximum potential energy of the dipole is 1.3 J .
3. Which of the following is true for the figure showing electric lines of force? (E is electrical field, V is potential)

- (a) $E_A > E_B$
 - (b) $E_B > E_A$
 - (c) $V_A < V_B$
 - (d) $V_B > V_A$


4. If we use permittivity ϵ , resistance R , gravitational constant G and voltage V as fundamental physical quantities, then

- (a) [angular displacement] $= \epsilon^0 R^0 G^0 V^0$
 - (c) [dipole moment] $= \epsilon^1 R^0 G^0 V^1$

- (b) [Velocity] $= \epsilon^{-1} R^{-1} G^0 V^0$
 - (d) [force] $= \epsilon^1 R^0 G^0 V^2$
5. Units of electric flux are

(a) $\frac{\text{N} \cdot \text{m}^2}{\text{Coul}^2}$

(b) $\frac{\text{N}}{\text{Coul}^2 \cdot \text{m}^2}$

(c) volt-m

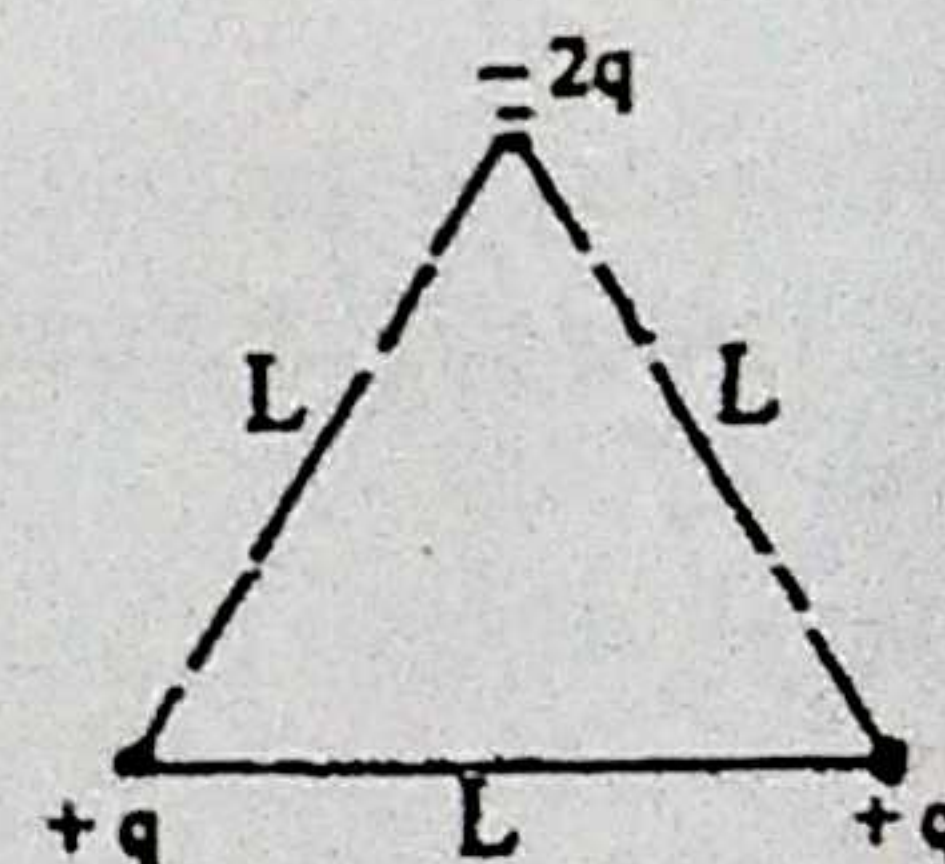
(d) volt-m³
6. Which of the following statements are correct?
 - (a) Electric field calculated by Gauss law is the field due to only those charges which are enclosed inside the Gaussian Surface.
 - (b) Gauss law is applicable only when there is a symmetrical distribution of charge.
 - (c) Electric flux through a closed surface will depend only on charges enclosed within that surface only.
 - (d) None of these
7. Mark the correct options:
 - (a) Gauss's law is valid only for uniform charge distributions.
 - (b) Gauss's law is valid only for charges placed in vacuum.
 - (c) The electric field calculated by Gauss's law is the field due to the charges.
 - (d) The flux of the electric field through a closed surface due to all the charges is equal to the flux due to the charges enclosed by the surface.

1. A thin-walled, spherical conducting shell S of radius R is given charge Q. The same amount of charge is also placed at its centre C. Which of the following statements are correct?
- On the outer surface of S, the charge density is $\frac{Q}{2\pi R^2}$
 - The electric field is zero at all the points inside S.
 - At a point just outside S, the electric field is double the field at a point just inside S.
 - At any points inside S, the electric field is inversely proportional to the square of its distance from C.

3. A hollow closed conductor irregular shape is given some charge. Which of the following statements are correct?
- The entire charge will appear on its outer surface.
 - All points on the conductor will have the same potential.
 - All points on the surface will have the same charge density.
 - All points near its surface and outside it will have the same electric intensity.

10. Charges Q_1 and Q_2 lie inside and outside respectively of a closed surface S. Let E be the field at any point on S and Φ be the flux of E over S.
- If Q_1 changes, both E and Φ will change.
 - If Q_2 changes, E will change but Φ will not change.
 - If Q_1 and $Q_2 \neq 0$ then $E \neq 0$ but $\Phi = 0$.
 - If $Q_1 \neq 0$ and $Q_2 = 0$ then $E = 0$ but $\Phi \neq 0$

11. Three point charges are placed at the corners of an equilateral triangle of side L as shown in the figure.

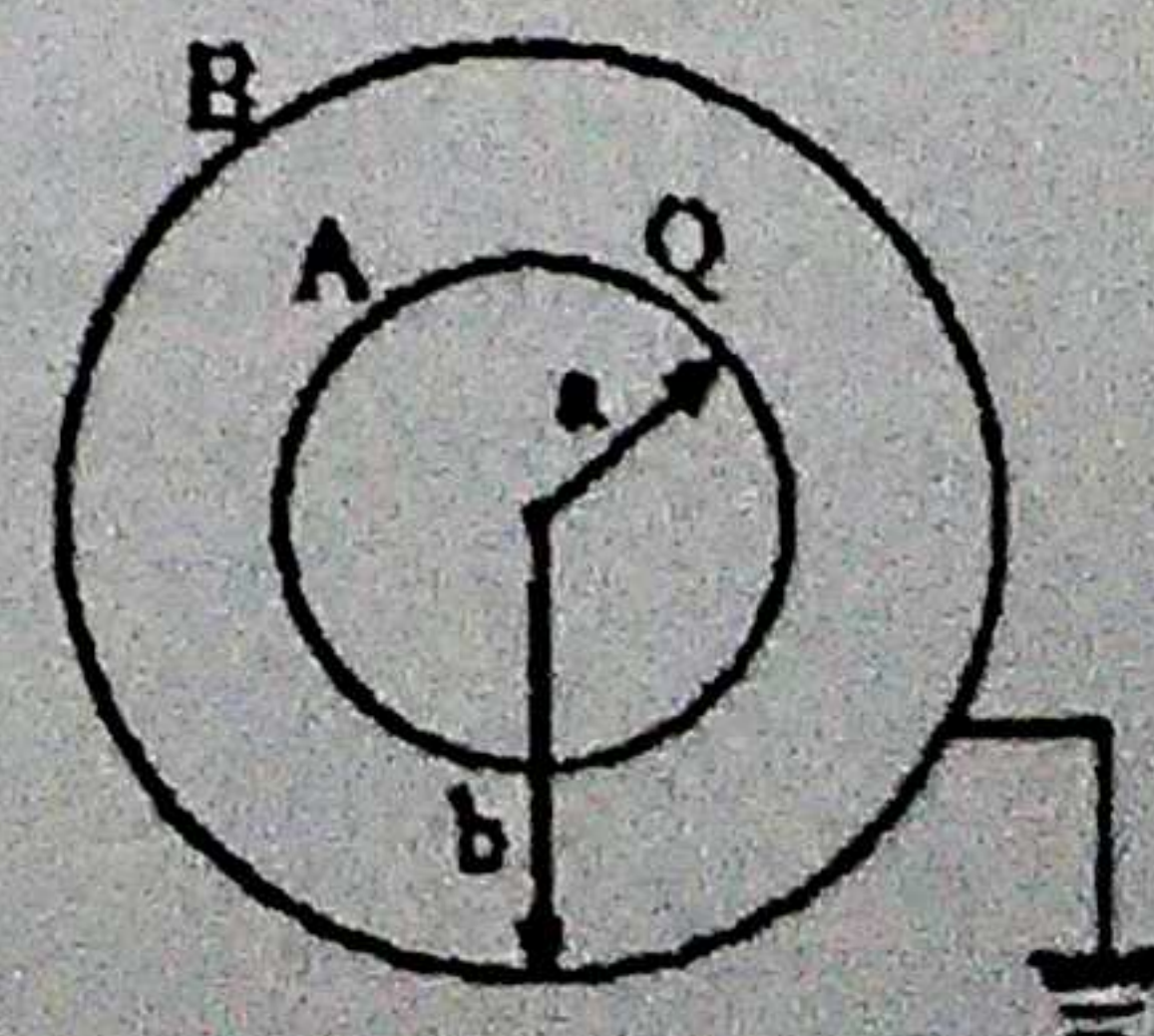


- The potential at the centroid of the triangle is zero.
 - The electric field at the centroid of the triangle is zero.
 - The dipole moment of the system is $\sqrt{2} qL$
 - The dipole moment of the system is $\sqrt{3} qL$
12. An electric dipole is placed at the centre of a sphere. Mark the correct answer
- the flux of the electric field through the sphere is zero
 - the electric field is zero at every point of the sphere
 - the electric potential is zero everywhere on the sphere
 - the electric potential is zero on a circle on the surface

13. An electric field converges at the origin whose magnitude is given by the expression $E = 100r \text{ Nt/Coul}$, where r is the distance measured from the origin.
- Total charge contained in any spherical volume with its center at origin is negative.
 - Total charge contained in any spherical volume, irrespective with its center at origin is negative.
 - Total charge contained in any spherical volume of radius 3 cm with its centre at origin has magnitude $3 \times 10^{-13} \text{ C}$.
 - Total charge contained in any spherical volume of radius 3 cm with its centre at origin has magnitude $3 \times 10^{-9} \text{ Coul}$.

14. A conducting sphere A of radius a, with charge Q, is placed concentrically inside a conducting shell B of radius b. B is earthed. C is the common center of the A and B.

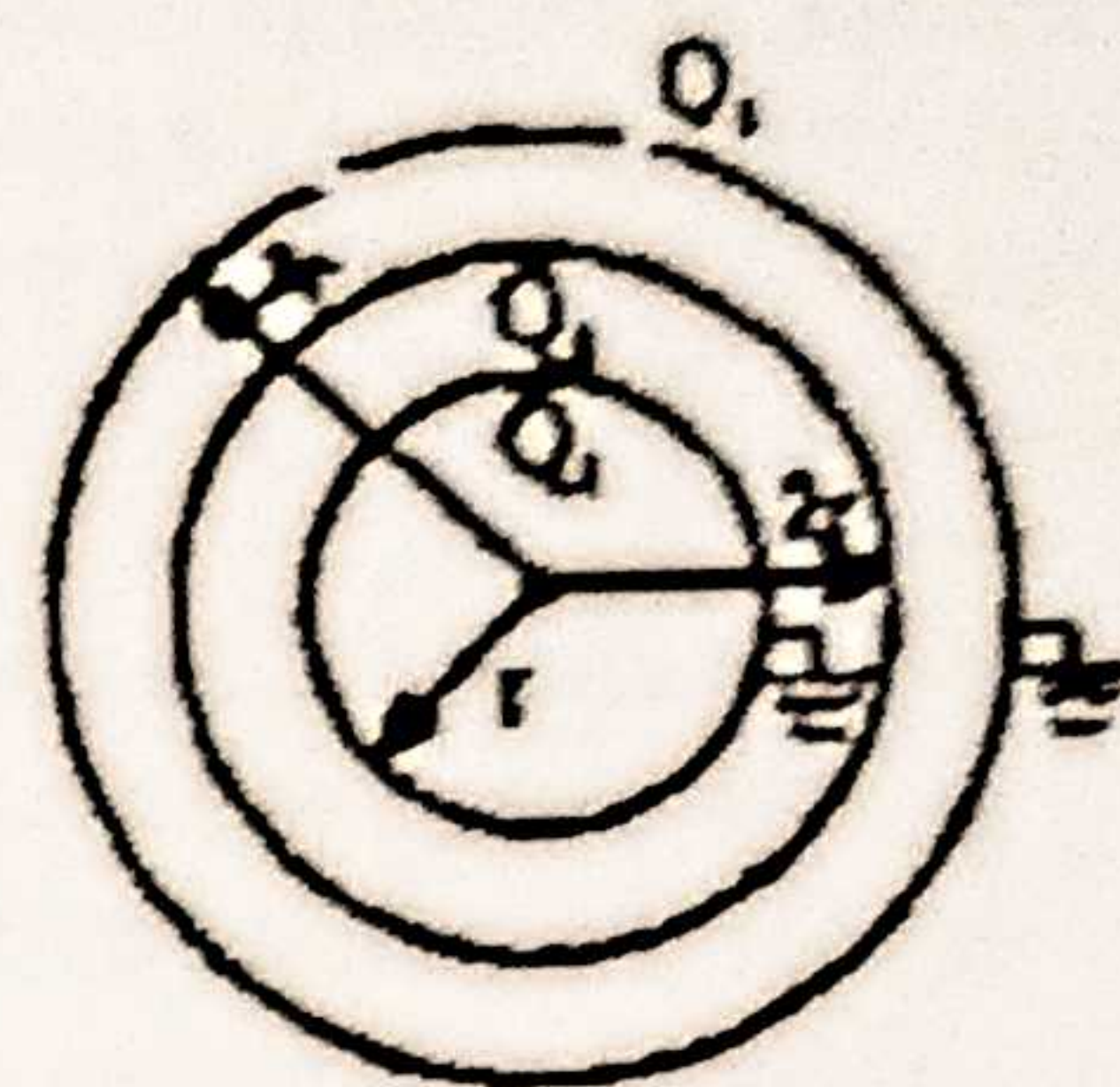
- The field is a distance r from C, where $a < r < b$ is $\frac{1}{4\pi\epsilon_0 r^2}$
- The potential at a distance r from C, where $a < r < b$, is $\frac{1}{4\pi\epsilon_0 r^2} \frac{Q}{r}$
- The potential difference between A and B is $\frac{1}{4\pi\epsilon_0 r^2} Q \left(\frac{1}{a} - \frac{1}{b} \right)$



(d) The potential at a distance r from C , where $a \leq r \leq b$, $\frac{1}{4\pi\epsilon_0 r^2} Q \left(\frac{1}{a} - \frac{1}{b} \right)$

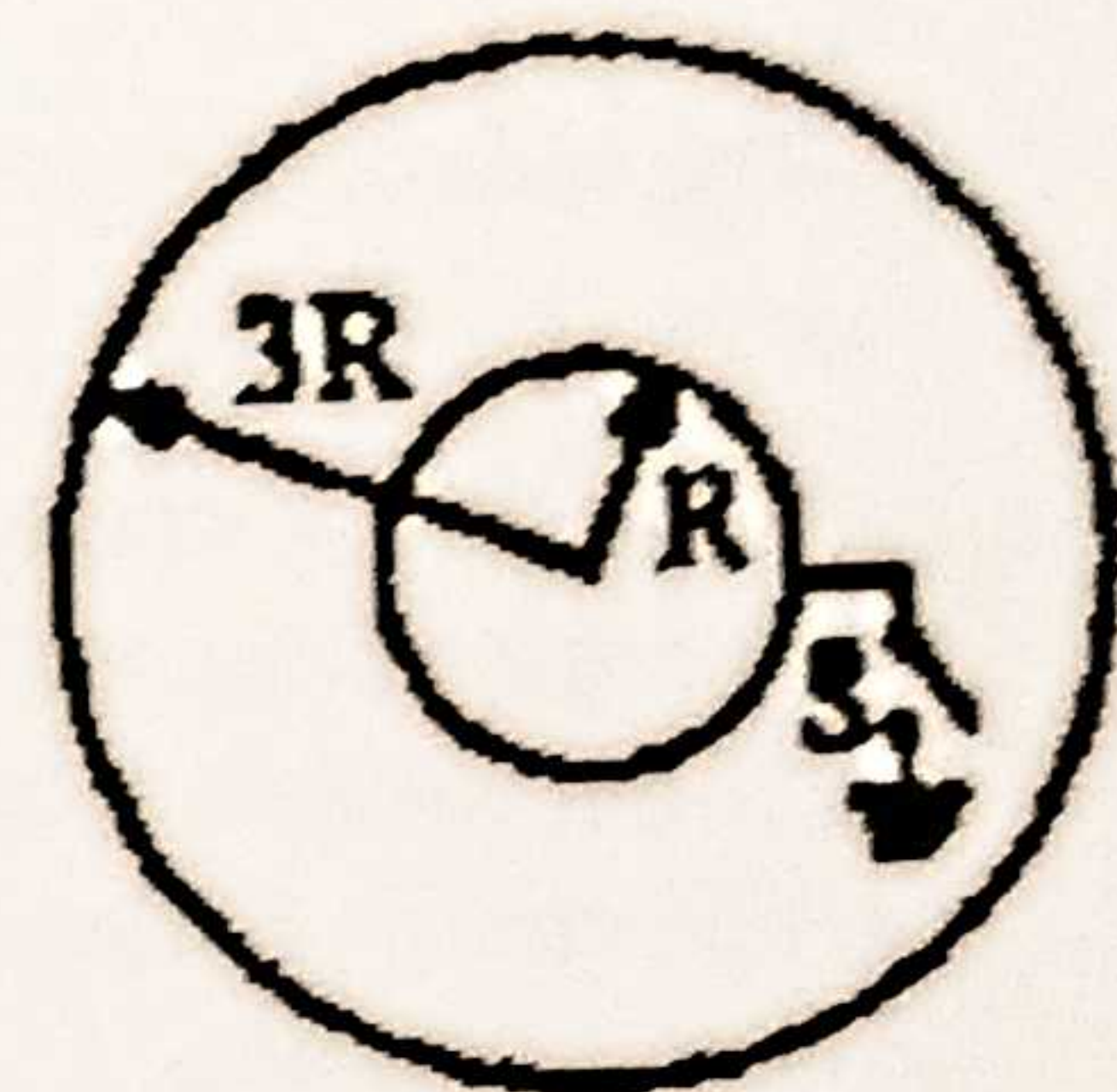
15. Three concentric conducting spherical shells have radius r , $2r$ and $3r$ and Q_1 , Q_2 and Q_3 are final charges respectively. Innermost and outer most shells are already earthed as shown in figure. Choose the wrong statement.

- (a) $Q_1 + Q_3 = -Q_2$ (b) $Q_1 = \frac{-Q_2}{4}$
 (c) $\frac{Q_3}{Q_1} = 3$ (d) $\frac{Q_3}{Q_2} = \frac{-1}{3}$



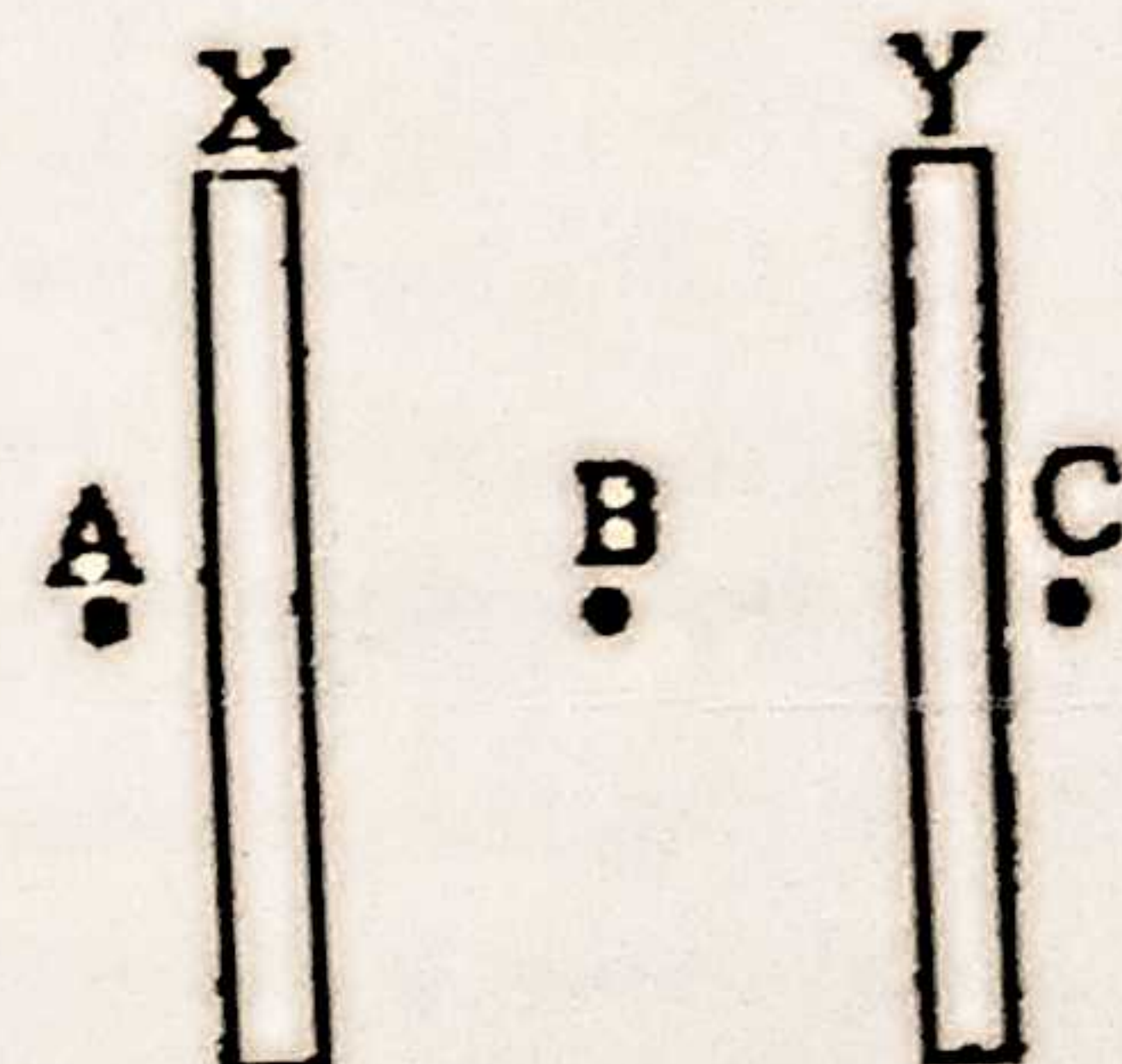
16. Two thin conducting shells of radii R and $3R$ are shown in the figure. The outer shell carries charge $+Q$ and the inner shell is neutral. The inner shell is earthed with the help of a switch S .

- (a) With the switch S open, the potential of the inner sphere is equal to that of the outer.
 (b) When the switch S is closed, the potential of the inner sphere becomes zero.
 (c) With the switch S closed, the charge attained by the inner sphere is $-q/3$.
 (d) By closing the switch the capacitance of the system increases.



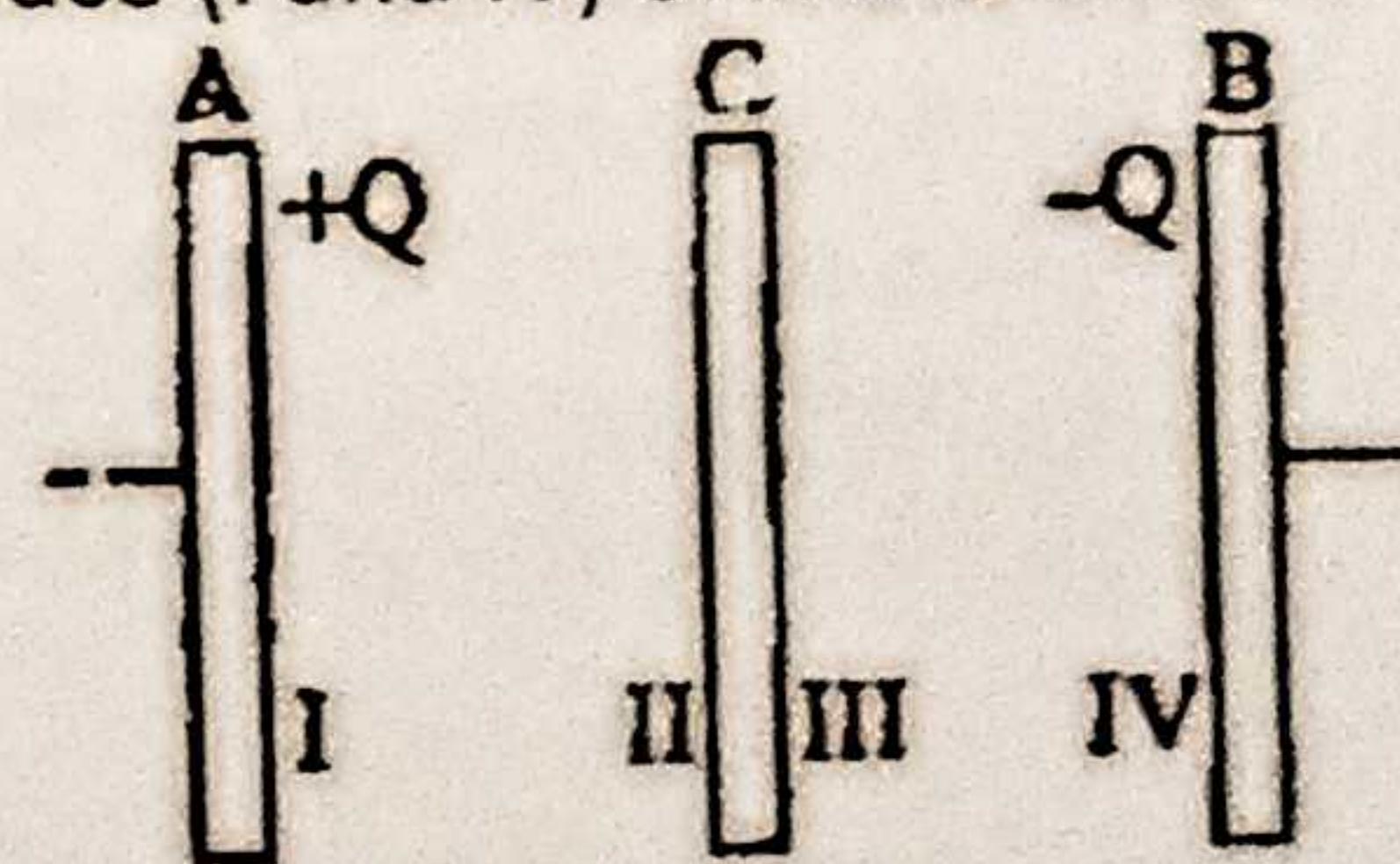
17. X and Y are large, parallel conducting plates closed to each other. Each face has an area A . X is given a charge Q . Y is without any charge. Points A , B and C are as shown in figure.

- (a) The field at B is $\frac{Q}{2\epsilon_0 A}$
 (b) The field at B is $\frac{Q}{\epsilon_0 A}$
 (c) The field at A , B and C are of the same magnitude.
 (d) The field at A and C are of the same magnitude, but in opposite directions.



18. Plates A and B constitute an isolated, charge parallel-plate capacitor. The inner surface (I and IV) of A and B have charges $+Q$ and $-Q$ respectively. A third plate C with charge $+Q$ is now introduced midway between A and B . Which of the following statements is not correct?

- (a) the surface I and II will have equal and opposite charges.
 (b) The surface III and IV will have equal and opposite charges.
 (c) the charge on surface III will be greater than Q .
 (d) the potential difference between A and C will be equal to the potential difference between C and B .

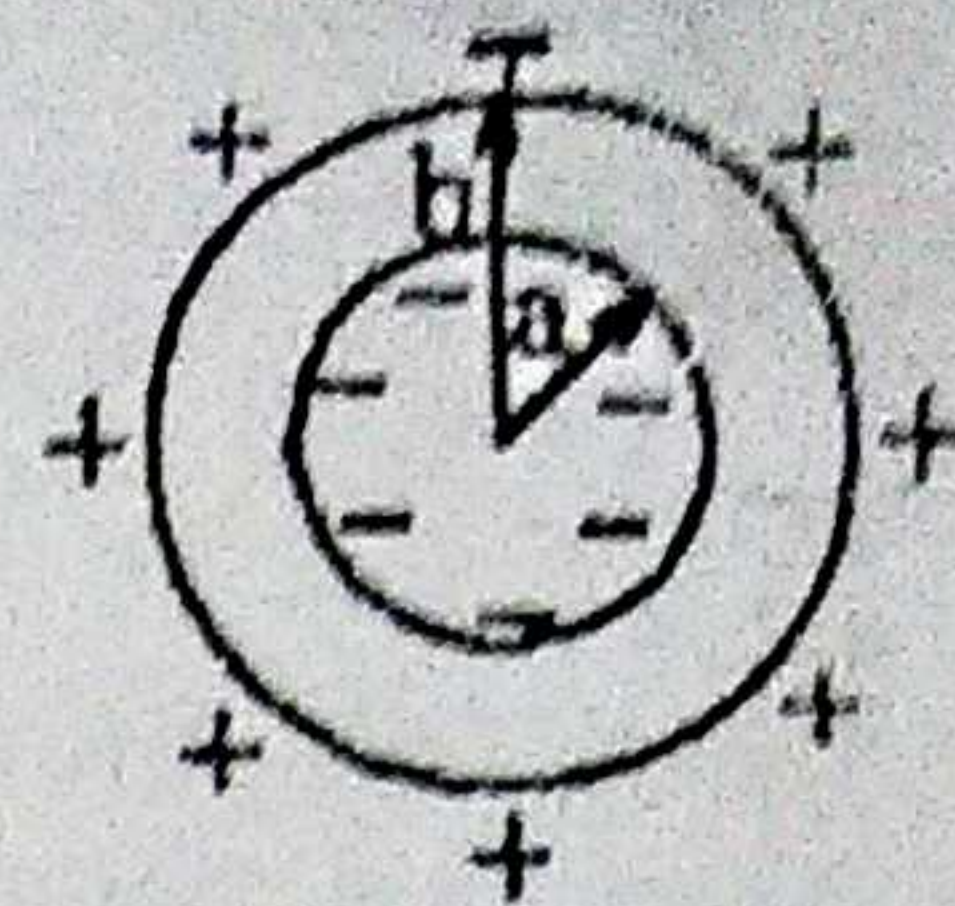


19. A particle of mass m and charge q is thrown in a region where uniform gravitational field and electric field are present. The path of particle is

- (a) may be straight line (b) may be a circle
 (c) may be a parabola (d) may be a hyperbola

Question No. (20 – 25) (6 – Questions)

An empty thick conducting shell of inner radius a and outer radius b is shown in figure. If it is observed that the inner face of the shell carries a uniform charge density $-\sigma$ and the surface carries a uniform charge density $+\sigma$



20. If a point charge q_A is placed at the centre of the shell, then choose the correct statement(s)
 - (a) The charge must be positive
 - (b) The charge must be negative
 - (c) The magnitude of charge must be $4\pi\sigma a^2$
 - (d) The magnitude of charge must be $4\pi\sigma(a^2 - b^2)$
21. If another point charge q_B is also placed at a distance $c(>b)$ the centre of shell, then choose the correct statements.
 - (a) force experienced by charge A is $\frac{\sigma q_A b^2}{\epsilon_0 c^2}$
 - (b) force experienced by charge A is zero
 - (c) The force experience by charge B is $\frac{\sigma q_B b^2}{\epsilon_0 c^2}$
 - (d) The force experienced by charge B is $\frac{kq_A q_B}{c^2}$
22. If the charge q_A is slowly moved inside the shell, then choose the statement(s)
 - (a) Charge distribution on the inner and outer face of the shell changes
 - (b) The force acting on the charge B changes
 - (c) The charge B also starts moving slowly
 - (d) None of these
23. Choose the correct statements related to the potential of the shell in absence of q_B
 - (a) Potential of the outer surface is more than that of the inner surface because it is positively charged
 - (b) Potential of the outer surface is more than that of the inner surface because it carries more charge
 - (c) Both the surfaces have equal potential
 - (d) The potential of the outer surface is $\frac{\sigma b}{\epsilon_0}$
24. If the outer surface of the shell is earthed, then identify the correct statement(s)
 - (a) Only the potential of outer surface becomes zero
 - (b) Charge on the outer surface also becomes zero
 - (c) The outer surface attains negative charge
 - (d) Negative charge on the inner surface decreases
25. If the inner surface of the shell is earthed, then identify the correct statement(s)
 - (a) the potential of the both inner and outer surface of the shell becomes zero
 - (b) Charge on the outer surface becomes zero
 - (c) Charge on the inner surface decreases
 - (d) Positive charge flows from the shell to the earth